



МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ
Федеральное государственное бюджетное образовательное учреждение
высшего образования
"МИРЭА - Российский технологический университет"

РТУ МИРЭА

Институт информационных технологий (ИТ)
Кафедра математического обеспечения и стандартизации информационных
технологий (МОСИТ)

ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ №3
по дисциплине
«Распределенные СУБД»

Тема. TTL & Tombstones.

Выполнил студент группы ИКБО-66-23

Смирнов А.Ю.

Принял преподаватель

Гуличева А.А.

Москва 2026

Практическая часть

1. Утилита sstabledump и ее установка

```
C:\Users\Redmi>docker pull cassandra
Using default tag: latest
latest: Pulling from library/cassandra
b5ce294f2c05: Pull complete
e5a5c7d9e1c6: Pull complete
e1dddef10ea4: Pull complete
c23f0b6a6c31: Pull complete
5936ae2ca844: Pull complete
f883331bfddb: Pull complete
a55097fdf8ad: Pull complete
6db0909c4473: Pull complete
8ed48d995d71: Download complete
03772bf4d7b1: Download complete
Digest: sha256:b494afb347bfd20765d80817d89d9e6e31ae77a339bf34e82c6ea868a51c9c1
Status: Downloaded newer image for cassandra:latest
docker.io/library/cassandra:latest
```

Н

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# echo "deb https://downloads.apache.org/cassandra/debian 40x main" | tee -a /etc/apt/sources.list.d/cassandra.sources.list
deb https://downloads.apache.org/cassandra/debian 40x main
root@82ccc61ea929:/# apt update && apt upgrade
Ign:1 https://downloads.apache.org/cassandra/debian 40x InRelease
Get:2 http://deb.debian.org/debian bookworm InRelease [121 kB]
Get:3 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:4 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:5 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
Get:6 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:7 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [2994 kB]
Ign:1 https://downloads.apache.org/cassandra/debian 40x InRelease
Ign:1 https://downloads.apache.org/cassandra/debian 40x InRelease
Err:1 https://downloads.apache.org/cassandra/debian 40x InRelease
  Certificate verification failed: The certificate is NOT trusted. The certificate issuer is unknown. Could not handshake: Error in the certificate verification. [IP: 135.181.214.104 443]
Fetched 3940 kB in 7s (5272 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: https://downloads.apache.org/cassandra/debian/dists/40x/InRelease: No system certificates available. Try installing ca-certificates.
W: https://downloads.apache.org/cassandra/debian/dists/40x/InRelease: No system certificates available. Try installing ca-certificates.
W: https://downloads.apache.org/cassandra/debian/dists/40x/InRelease: No system certificates available. Try installing ca-certificates.
W: https://downloads.apache.org/cassandra/debian/dists/40x/InRelease: No system certificates available. Try installing ca-certificates.
W: Failed to fetch https://downloads.apache.org/cassandra/debian/dists/40x/InRelease Certificate verification failed: The certificate is NOT trusted. The certificate issuer is unknown. Could not handshake: Error in the certificate verification. [IP: 135.181.214.104 443]
W: Some index files failed to download. They have been ignored, or old ones used instead.
```

h

```
root@82ccc61ea929:/# apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.88.1-10+deb12u4).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@82ccc61ea929:/# apt-get install gnupg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  dirnrg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm libksba8 libnpth0 pinentry-curses
Suggested packages:
  dbus-user-session libpam-systemd pinentry-gnome3 tor parcimonie xloadimage scdaemon pinentry-doc
The following NEW packages will be installed:
  dirnrg gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm libksba8 libnpth0 pinentry-curses
0 upgraded, 14 newly installed, 0 to remove and 0 not upgraded.
Need to get 7885 kB of archives.
After this operation, 16.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main amd64 libksba8 amd64 2.5.5-5 [48.5 kB]
Get:2 http://deb.debian.org/debian bookworm/main amd64 gpgconf amd64 2.2.40-1.1+deb12u2 [565 kB]
Get:3 http://deb.debian.org/debian bookworm/main amd64 libksba8 amd64 1.6.3-2 [128 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 libnpth0 amd64 1.6-2 [19.0 kB]
Get:5 http://deb.debian.org/debian bookworm/main amd64 dirnrg amd64 2.2.40-1.1+deb12u2 [793 kB]
Get:6 http://deb.debian.org/debian bookworm/main amd64 gnupg-l10n all 2.2.40-1.1+deb12u2 [1093 kB]
Get:7 http://deb.debian.org/debian bookworm/main amd64 gnupg-utils amd64 2.2.40-1.1+deb12u2 [927 kB]
Get:8 http://deb.debian.org/debian bookworm/main amd64 gpg amd64 2.2.40-1.1+deb12u2 [950 kB]
Get:9 http://deb.debian.org/debian bookworm/main amd64 pinentry-curses amd64 1.2.1-1 [77.4 kB]
Get:10 http://deb.debian.org/debian bookworm/main amd64 gpg-agent amd64 2.2.40-1.1+deb12u2 [693 kB]
Get:11 http://deb.debian.org/debian bookworm/main amd64 gpg-wks-client amd64 2.2.40-1.1+deb12u2 [541 kB]
```

```
Processing triggers for libc-bin (2.36-9) ...
root@82ccc61ea929:/# curl https://downloads.apache.org/cassandra/KEYS | apt-key add -
% Total % Received % Xferd Average Speed Time Time Time Current
0 0 0 0 0 0 0 0 0 0 0 0 Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
100 274k 100 274k 0 0 1197k 0 0 0 0 0 1190k
OK
root@82ccc61ea929:/# apt-get update
Ign:1 https://downloads.apache.org/cassandra/debian 40x InRelease
Hit:2 http://deb.debian.org/debian bookworm InRelease
Err:3 https://downloads.apache.org/cassandra/debian 40x Release
  404 Not Found [IP: 135.181.214.104 443]
Hit:4 http://deb.debian.org/debian bookworm-updates InRelease
Hit:5 http://deb.debian.org/debian-security bookworm-security InRelease
Reading package lists... Done
E: The repository 'https://downloads.apache.org/cassandra/debian 40x Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
root@82ccc61ea929:/# apt-get install cassandra-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# apt-get install cassandra-tools -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package cassandra-tools
root@82ccc61ea929:/# |
```

2. Создание keyspaces и таблиц

```
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE cycling WITH replication =
... {'class': 'SimpleStrategy', 'replication_factor': '1'} AND durable_writes =
... true;
cqlsh> DESCRIBE KEYSPACE
No keyspace specified and no current keyspace
cqlsh> DESCRIBE KEYSPACES

cycling system_auth system_schema system_views
system system_distributed system_traces system_virtual_schema

cqlsh> |
```

```
cqlsh> CREATE TABLE cycling.rank_by_year_and_name (
... race_year int,
... race_name text,
... rank int,
... cyclist_name text,
... PRIMARY KEY ((race_year, race_name), rank)
... ) WITH CLUSTERING ORDER BY (rank ASC);
cqlsh> |
```

```
cqlsh> DESC TABLES

Keyspace cycling
-----
rank_by_year_and_name
```

```
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

race_year | race_name | rank | cyclist_name
-----|-----|-----|-----
2014 | 4th Tour of Beijing | 1 | Phillippe GILBERT
2014 | 4th Tour of Beijing | 2 | Daniel MARTIN
2014 | 4th Tour of Beijing | 3 | Johan Esteban CHAVES
2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN
2015 | Giro d'Italia - Stage 11 - Forli > Imola | 2 | Carlos BETANCUR
2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PRADES
2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 2 | Adam PHELAN
2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 3 | Thomas LEBAS

(8 rows)
cqlsh> |
```

```
cqlsh> DESC TABLES

Keyspace cycling
-----
calendar  rank_by_year_and_name
```

3 Выгрузка в Sstable

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# nodetool flush cycling;
root@82ccc61ea929:/#
```

```
root@82ccc61ea929:/# ls /var/lib/cassandra/data/cycling
calendar-9e46ece022fb11f1b91c0560ca0844e3  rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3#
```

```
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3# cat >> /etc/cassandra/jvm-clients.options << EOF
> --add-opens java.base/java.io=ALL-UNNAMED
> --add-opens java.base/java.nio=ALL-UNNAMED
> --add-opens java.base/sun.nio.ch=ALL-UNNAMED
> --add-opens java.base/java.lang=ALL-UNNAMED
> EOF
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3# cat /etc/cassandra/jvm-clients.options
--add-opens java.base/java.io=ALL-UNNAMED
--add-opens java.base/java.nio=ALL-UNNAMED
--add-opens java.base/sun.nio.ch=ALL-UNNAMED
--add-opens java.base/java.lang=ALL-UNNAMED
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fa11f1b91c0560ca0844e3# sstabledump nb-1-big-Data.db
[
  {
    "partition" : {
      "key" : [ "201u", "4th Tour of Beijing" ],
      "position" : 0
    },
    "rows" : [
      {
        "type" : "row",
        "position" : 43,
        "clustering" : [ 1 ],
        "liveness_info" : { "tstamp" : "2026-03-18T18:50:04.286584Z" },
        "cells" : [
          { "name" : "cyclist_name", "value" : "Phillippe GILBERT" }
        ]
      },
      {
        "type" : "row",
        "position" : 74,
        "clustering" : [ 2 ],
        "liveness_info" : { "tstamp" : "2026-03-18T18:50:17.752584Z" },
        "cells" : [
          { "name" : "cyclist_name", "value" : "Daniel MARTIN" }
        ]
      },
      {
        "type" : "row",
        "position" : 181,
        "clustering" : [ 3 ],
        "liveness_info" : { "tstamp" : "2026-03-18T18:50:39.122719Z" },
        "cells" : [
          { "name" : "cyclist_name", "value" : "Johan Esteban CHAVES" }
        ]
      }
    ]
  },
  {
    "partition" : {
      "key" : [ "2015", "Giro d'Italia - Stage 11 - Forlì > Imola" ],
      "position" : 137
    },
    "rows" : [
      {
        "type" : "row",
        "position" : 281,
```

4 Tombstone для разделов

```
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> DELETE from cycling.rank_by_year_and_name WHERE race_year = 2014 AND race_name = '4th Tour of Beijing';
cqlsh> |
```

```
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

+-----+-----+-----+-----+
| race_year | race_name | rank | cyclist_name |
+-----+-----+-----+-----+
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 2 | Carlos BETANCUR |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PRADES |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 2 | Adam PHELAN |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 3 | Thomas LEBAS |
+-----+-----+-----+-----+

(5 rows)
cqlsh> |
```

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db  nb-1-big-Index.db  nb-2-big-CompressionInfo.db  nb-2-big-Index.db
nb-1-big-Data.db  nb-1-big-Statistics.db  nb-2-big-Data.db  nb-2-big-Statistics.db
nb-1-big-Digest.crc32  nb-1-big-Summary.db  nb-2-big-Digest.crc32  nb-2-big-Summary.db
nb-1-big-Filter.db  nb-1-big-TOC.txt  nb-2-big-Filter.db  nb-2-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# sstabledump nb-2-big-Data.db
[
  {
    "partition" : {
      "key" : [ "2014", "4th Tour of Beijing" ],
      "position" : 0,
      "deletion_info" : { "marked_deleted" : "2026-03-18T19:47:21.090869Z", "local_delete_time" : "2026-03-18T19:47:21Z" }
    },
    "rows" : [ ]
  }
]
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# |
```

5 Tombstone для строк

```
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> DELETE FROM cycling.rank_by_year_and_name WHERE race_year = 2015 AND race_name = 'Giro d'Italia - Stage 11 - Forli > Imola' AND rank = 2;
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

+-----+-----+-----+-----+
| race_year | race_name | rank | cyclist_name |
+-----+-----+-----+-----+
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PRADES |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 2 | Adam PHELAN |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 3 | Thomas LEBAS |
+-----+-----+-----+-----+

(4 rows)
cqlsh> |
```

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db  nb-1-big-Summary.db  nb-2-big-Index.db  nb-3-big-Digest.crc32
nb-1-big-Data.db  nb-1-big-TOC.txt  nb-2-big-Statistics.db  nb-3-big-Filter.db
nb-1-big-Digest.crc32  nb-2-big-CompressionInfo.db  nb-2-big-Summary.db  nb-3-big-Index.db
nb-1-big-Filter.db  nb-2-big-Data.db  nb-2-big-TOC.txt  nb-3-big-Statistics.db
nb-1-big-Index.db  nb-2-big-Digest.crc32  nb-3-big-CompressionInfo.db  nb-3-big-Summary.db
nb-1-big-Statistics.db  nb-2-big-Filter.db  nb-3-big-Data.db  nb-3-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# sstabledump nb-3-big-Data.db
[
  {
    "partition" : {
      "key" : [ "2015", "Giro d'Italia - Stage 11 - Forli > Imola" ],
      "position" : 0
    },
    "rows" : [
      {
        "type" : "row",
        "position" : 64,
        "clustering" : [ 2 ],
        "deletion_info" : { "marked_deleted" : "2026-03-18T19:55:43.525636Z", "local_delete_time" : "2026-03-18T19:55:43Z" },
        "cells" : [ ]
      }
    ]
  }
]
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-56fa792022fall1fb91c0560ca0844e3# |
```

6 Tombstone для диапазонов

```
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> DROP TABLE cycling.rank_by_year_and_name;
cqlsh> DESC TABLES

Keyspace cycling
-----
calendar
```

```
cqlsh> CREATE TABLE cycling.rank_by_year_and_name (
... race_year int,
... race_name text,
... rank int,
... cyclist_name text,
... PRIMARY KEY ((race_year, race_name), rank)
... ) WITH CLUSTERING ORDER BY (rank ASC);
```

```
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2015, 'Tour of Japan - Stage 4 - Minami > Shinshu', 'Benjamin PARADES', 1);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2015, 'Tour of Japan - Stage 4 - Minami > Shinshu', 'Adam PHELAN', 2);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2015, 'Tour of Japan - Stage 4 - Minami > Shinshu', 'Thomas LEBAS', 3);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2015, 'Giro d'Italia - Stage 11 - Forli > Imola', 'Ilnur ZAKARIN', 1);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2015, 'Giro d'Italia - Stage 11 - Forli > Imola', 'Carlos BETANCUR', 2);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2014, '4th Tour of Beijing', 'Phillippe GILBERT', 1);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2014, '4th Tour of Beijing', 'Daniel MARTIN', 2);
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2014, '4th Tour of Beijing', 'Johan Esteban CHAVES', 3);
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

+-----+-----+-----+-----+
| race_year | race_name | rank | cyclist_name |
+-----+-----+-----+-----+
| 2014 | 4th Tour of Beijing | 1 | Phillippe GILBERT |
| 2014 | 4th Tour of Beijing | 2 | Daniel MARTIN |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 2 | Carlos BETANCUR |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PARADES |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 2 | Adam PHELAN |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 3 | Thomas LEBAS |
+-----+-----+-----+-----+
(7 rows)
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2014, '4th Tour of Beijing', 'Johan Esteban CHAVES', 3);
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;
```

```
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank) VALUES (2014, '4th Tour of Beijing', 'Johan Esteban CHAVES', 3);
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

+-----+-----+-----+-----+
| race_year | race_name | rank | cyclist_name |
+-----+-----+-----+-----+
| 2014 | 4th Tour of Beijing | 1 | Phillippe GILBERT |
| 2014 | 4th Tour of Beijing | 2 | Daniel MARTIN |
| 2014 | 4th Tour of Beijing | 3 | Johan Esteban CHAVES |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 2 | Carlos BETANCUR |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PARADES |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 2 | Adam PHELAN |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 3 | Thomas LEBAS |
+-----+-----+-----+-----+
(8 rows)
cqlsh> |
```

```
cqlsh> DELETE from cycling.rank_by_year_and_name WHERE race_year = 2015 AND
... race_name = 'Tour of Japan - Stage 4 - Minami > Shinshu' AND rank > 1;
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

+-----+-----+-----+-----+
| race_year | race_name | rank | cyclist_name |
+-----+-----+-----+-----+
| 2014 | 4th Tour of Beijing | 1 | Phillippe GILBERT |
| 2014 | 4th Tour of Beijing | 2 | Daniel MARTIN |
| 2014 | 4th Tour of Beijing | 3 | Johan Esteban CHAVES |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 1 | Ilnur ZAKARIN |
| 2015 | Giro d'Italia - Stage 11 - Forli > Imola | 2 | Carlos BETANCUR |
| 2015 | Tour of Japan - Stage 4 - Minami > Shinshu | 1 | Benjamin PARADES |
+-----+-----+-----+-----+
(6 rows)
cqlsh> |
```

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/
root@82ccc61ea929:/var/lib/cassandra/data/cycling# ls
calendar-9e46ce022fb11fb91c0560ca0844e3 rank_by_year_and_name-81c2c530230511fb91c0560ca0844e3
rank_by_year_and_name-56fa792022fa11fb91c0560ca0844e3
```

```
root@82ccc61ea929:/var/lib/cassandra/data/cycling# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling# cd /var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511fb91c0560ca0844e3/
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511fb91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db nb-1-big-Digest.crc32 nb-1-big-Index.db nb-1-big-Summary.db
nb-1-big-Data.db nb-1-big-Filter.db nb-1-big-Statistics.db nb-1-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511fb91c0560ca0844e3# |
```

```

"partition" : {
  "key" : [ "2015", "Tour of Japan - Stage 4 - Minami > Shinshu" ],
  "position" : 259
},
"rows" : [
  {
    "type" : "row",
    "position" : 325,
    "clustering" : [ 1 ],
    "liveness_info" : { "tstamp" : "2026-03-18T20:05:11.059388Z" },
    "cells" : [
      { "name" : "cyclist_name", "value" : "Benjamin PARADES" }
    ]
  },
  {
    "type" : "range_tombstone_bound",
    "start" : {
      "type" : "exclusive",
      "clustering" : [ 1 ],
      "deletion_info" : { "marked_deleted" : "2026-03-18T20:15:02.760419Z", "local_delete_time" : "2026-03-18T20:15:02Z" }
    },
    {
      "type" : "range_tombstone_bound",
      "end" : {
        "type" : "inclusive",
        "deletion_info" : { "marked_deleted" : "2026-03-18T20:15:02.760419Z", "local_delete_time" : "2026-03-18T20:15:02Z" }
      }
    }
  ]
}
}
}
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3#

```

7 Complex column Tombstones

```

cqlsh> CREATE TABLE cycling_cyclist_career_teams (
... id UUID PRIMARY KEY,
... lastname text,
... teams set<text>
... );
cqlsh> INSERT INTO cycling_cyclist_career_teams (id, lastname, teams) VALUES (cb97baad-eac8-4f65-b28a-bddc06a0de23, 'ARMITSTEAD', {'Boels-Dolmans Cycling Team', 'AA Drink - Leontien.nl', 'Team Garmin - Cervelo'});
cqlsh> SELECT * FROM cycling_cyclist_career_teams;

```

id	lastname	teams
cb97baad-eac8-4f65-b28a-bddc06a0de23	ARMITSTEAD	{ 'AA Drink - Leontien.nl', 'Boels-Dolmans Cycling Team', 'Team Garmin - Cervelo' }

```

(1 rows)
cqlsh>

```

```

C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/
root@82ccc61ea929:/var/lib/cassandra/data/cycling# ls
calendar-9e46cece022fb1f1b91c0560ca0844e3  rank_by_year_and_name-56fa792022f11f1b91c0560ca0844e3
cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3  rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling# cd /var/lib/cassandra/data/cycling/cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling/cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling/cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db  nb-1-big-Digest_crc32  nb-1-big-Index.db  nb-1-big-Summary.db
nb-1-big-Data.db            nb-1-big-Filter.db    nb-1-big-Statistics.db  nb-1-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3# sstabledump nb-1-big-Data.db
[
  {
    "partition" : {
      "key" : [ "cb07baad-eac8-4f65-b28a-bddc06a0de23" ],
      "position" : 0
    },
    "rows" : [
      {
        "type" : "row",
        "position" : 30,
        "liveness_info" : { "tstamp" : "2026-03-18T20:34:08.635387Z" },
        "cells" : [
          { "name" : "lastname", "value" : "ARMITSTEAD" },
          { "name" : "teams", "deletion_info" : { "marked_deleted" : "2026-03-18T20:34:08.635386Z", "local_delete_time" : "2026-03-18T20:34:08Z" } },
          { "name" : "teams", "path" : [ "AA Drink - Leontien.nl" ], "value" : "" },
          { "name" : "teams", "path" : [ "Boels-Dolmans Cycling Team" ], "value" : "" },
          { "name" : "teams", "path" : [ "Team Garmin - Cervelo" ], "value" : "" }
        ]
      }
    ]
  }
]
root@82ccc61ea929:/var/lib/cassandra/data/cycling/cyclist_career_teams-c89a4fb0230911f1b91c0560ca0844e3#

```

8 Tombstone для ячейки

```

C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> INSERT INTO cycling.rank_by_year_and_name (race_year, race_name, cyclist_name, rank)
... VALUES (2018, 'Giro d'Italia - Stage 11 - Osimo > Imola', null, 1);
cqlsh> SELECT * FROM cycling.rank_by_year_and_name;

```

race_year	race_name	rank	cyclist_name
2018	Giro d'Italia - Stage 11 - Osimo > Imola	1	null
2014	4th Tour of Beijing	1	Phillippe GILBERT
2014	4th Tour of Beijing	2	Daniel MARTIN
2014	4th Tour of Beijing	3	Johan Esteban CHAVES
2015	Giro d'Italia - Stage 11 - Forli > Imola	1	Ilnur ZAKARIN
2015	Giro d'Italia - Stage 11 - Forli > Imola	2	Carlos BETANCUR
2015	Tour of Japan - Stage 4 - Minami > Shinshu	1	Benjamin PARADES

```

(7 rows)
cqlsh>

```

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# /var/lib/cassandra/data/cycling
bash: /var/lib/cassandra/data/cycling: Is a directory
root@82ccc61ea929:/# /var/lib/cassandra/data/cycling/
bash: /var/lib/cassandra/data/cycling/: Is a directory
root@82ccc61ea929:/# /var/lib/cassandra/data/cycling/
bash: /var/lib/cassandra/data/cycling/: Is a directory
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling
root@82ccc61ea929:/var/lib/cassandra/data/cycling# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling# cd rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db  nb-1-big-Index.db  nb-2-big-CompressionInfo.db  nb-2-big-Index.db
nb-1-big-Data.db  nb-1-big-Statistics.db  nb-2-big-Data.db  nb-2-big-Statistics.db
nb-1-big-Digest_crc32  nb-1-big-Summary.db  nb-2-big-Digest_crc32  nb-2-big-Summary.db
nb-1-big-Filter.db  nb-1-big-TOC.txt  nb-2-big-Filter.db  nb-2-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3# sstabledump nb-3-big-Data.db
Cannot find file /var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3/nb-3-big-Data.db
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3# sstabledump nb-2-big-Data.db
{
  "partition" : {
    "key" : [ "2018", "Giro d'Italia - Stage 11 - Osimo > Inola" ],
    "position" : 0
  },
  "rows" : [
    {
      "type" : "row",
      "position" : 64,
      "clustering" : [ 1 ],
      "liveness_info" : { "tstamp" : "2026-03-18T20:42:03.110468Z" },
      "cells" : [
        {
          "name" : "cyclist_name", "deletion_info" : { "local_delete_time" : "2026-03-18T20:42:03Z" }
        }
      ]
    }
  ]
}
```

9. Установка времени жизни записи TTL

```
C:\Users\Redmi>docker exec -ti cas cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.6 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> INSERT INTO cycling.calendar (race_id, race_name, race_start_date, race_end_date) VALUES (200, 'placeholder', '2015-05-27', '2015-05-27') USING TTL 86400;
cqlsh> SELECT * FROM cycling.calendar;

  race_id | race_start_date | race_end_date | race_name
-----|-----|-----|-----
    200 | 2015-05-27 00:00:00.000000+0000 | 2015-05-27 00:00:00.000000+0000 | placeholder

(1 rows)
cqlsh> |
```

```
cqlsh> UPDATE cycling.calendar USING TTL 300 SET race_name = 'placeholder' WHERE race_id = 200 AND race_start_date = '2015-05-27' AND race_end_date = '2015-05-27';
cqlsh> SELECT TTL (race_name) from cycling.calendar WHERE race_id = 200;
SyntaxException: line 1:17 mismatched input 'name' expecting <FROM (SELECT TTL (race)[name]...)
cqlsh> SELECT TTL (race_name) from cycling.calendar WHERE race_id = 200;

  ttl(race_name)
-----
          255

(1 rows)
cqlsh> |
```

10 TTL tombstones

```
C:\Users\Redmi>docker exec -ti cas bash
root@82ccc61ea929:/# cd /var/lib/cassandra/data/cycling/
root@82ccc61ea929:/var/lib/cassandra/data/cycling# nodetool flush
root@82ccc61ea929:/var/lib/cassandra/data/cycling# ls
calendar-9e46ece022fb11f1b91c0560ca0844e3  rank_by_year_and_name-86fa792023f1f1b91c0560ca0844e3
cyclist_team-09a4f0b23091f1f1b91c0560ca0844e3  rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling# cd calendar-9e46ece022fb11f1b91c0560ca0844e3
root@82ccc61ea929:/var/lib/cassandra/data/cycling/calendar-9e46ece022fb11f1b91c0560ca0844e3# ls
nb-1-big-CompressionInfo.db  nb-1-big-Digest_crc32  nb-1-big-Index.db  nb-1-big-Summary.db
nb-1-big-Data.db  nb-1-big-Filter.db  nb-1-big-Statistics.db  nb-1-big-TOC.txt
root@82ccc61ea929:/var/lib/cassandra/data/cycling/calendar-9e46ece022fb11f1b91c0560ca0844e3# sstabledump nb-big-Data.db
Cannot find file /var/lib/cassandra/data/cycling/calendar-9e46ece022fb11f1b91c0560ca0844e3/nb-big-Data.db
root@82ccc61ea929:/var/lib/cassandra/data/cycling/calendar-9e46ece022fb11f1b91c0560ca0844e3# sstabledump nb-1-big-Data.db
{
  "partition" : {
    "key" : [ "200" ],
    "position" : 0
  },
  "rows" : [
    {
      "type" : "row",
      "position" : 38,
      "clustering" : [ "2015-05-27 00:00:00.000Z", "2015-05-27 00:00:00.000Z" ],
      "liveness_info" : { "tstamp" : "2026-03-18T21:02:14.186049Z", "ttl" : 86400, "expires_at" : "2026-03-19T21:02:14Z", "expired" : false },
      "cells" : [
        {
          "name" : "race_name", "value" : "placeholder", "tstamp" : "2026-03-18T21:04:25.034066Z", "ttl" : 300, "expires_at" : "2026-03-18T21:09:25Z", "expired" : true }
        ]
      ]
    }
  ]
}
```

```
root@82ccc61ea929:/var/lib/cassandra/data/cycling/rank_by_year_and_name-81c2c530230511f1b91c0560ca0844e3# sstabledump nb-2-big-Data.db
{
  "partition" : {
    "key" : [ "2018", "Giro d'Italia - Stage 11 - Osimo > Inola" ],
    "position" : 0
  },
  "rows" : [
    {
      "type" : "row",
      "position" : 64,
      "clustering" : [ 1 ],
      "liveness_info" : { "tstamp" : "2026-03-18T20:42:03.110468Z" },
      "cells" : [
        {
          "name" : "cyclist_name", "deletion_info" : { "local_delete_time" : "2026-03-18T20:42:03Z" }
        }
      ]
    }
  ]
}
```


Контрольные вопросы:

С точки зрения архитектуры распределенных систем, это **одноранговая (Peer-to-Peer)** распределенная база данных.

Почему именно к этому типу (исходя из нашей практической работы):

- **Отсутствие главного узла (Masterless / P2P):** В нашем кластере нет единого сервера-"хозяина" и серверов-"подчиненных". Все запущенные нами узлы (cas1, cas2) абсолютно равны. Именно поэтому вы могли подключиться ко второму узлу и увидеть данные, записанные через первый.
- **Высокая доступность и устойчивость к разделению (AP по CAP-теореме):** Мы специально разнесли узлы по разным виртуальным датацентрам (datacenter1 и datacenter2). Если один датацентр "упадет", кластер продолжит работу, предоставляя доступ к данным из другого.
- **Децентрализованная репликация данных:** При создании *Keyspace* мы использовали `NetworkTopologyStrategy` и явно указали, что хотим хранить по одной копии данных в каждом датацентре. Система сама прозрачно для нас копирует данные между серверами.

```
docker run -d --name <название> cassandra:latest
```